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Understanding Analysis Sections Covered: 7.1, 7.2

Definitions and Theorems to include in Analysis Journal:

- Definition 7.2.1 (partition)
 - Definition 7.2.2 (refinement)
 - Lemmas 7.2.3, 7.2.4, 7.2.6
 - Definition 7.2.5 (upper and lower integral)
 - Definition 7.2.7 (Riemann Integrability)
-

Before Class:

Readings: Sections 9.1-9.4 of Alcock. Make sure and keep good track of the definitions Alcock presents, using pictures.

Analysis Journal Prompts:

- What do you remember about the definition of Riemann Integrals from calculus? *my guess is you may not have gone over these definitions like with differentiability and continuity in Calc 3 or MAT 300.*
- Why do you think we need all of this technical mumbo-jumbo? What wasn't sufficient with our ideas before?
- Are all integrable functions differentiable? Vice versa? (Which is stronger?)

Discuss: *What assumptions are we making when we say that integrals = antiderivatives? What's the issue, analysis-wise, with that?*

A **partition** P of $[a, b]$ is...

A partition Q is a **refinement** of a partition P if...

For each subinterval $[x_{k-1}, x_k]$ of P , let

$$m_k =$$

$$M_k =$$

The **lower sum** of f with respect to P is

$$L(f, P) =$$

The **upper sum** of f with respect to P is

$$U(f, P) =$$

Observation: $U(f, P) \text{ — } L(f, P)$

If $P \subseteq Q$, then $L(f, P) \underline{\hspace{1cm}} L(f, Q)$ and $U(f, P) \underline{\hspace{1cm}} U(f, Q)$

Proof.

☐

Lemma 7.2.4

If P_1 and P_2 are any two partitions of $[a, b]$, then $L(f, P_1)$ _____ $U(f, P_2)$

Proof.

□

Upper Integral, Lower Integral

Let $\mathcal{P}$ be the collection of all possible partitions of $[a, b]$.

The **lower integral** of f is

$$L(f) =$$

The **upper sum** of f is

$$U(f) =$$

Lemma 7.2.6

For a bounded function f defined on $[a, b]$, $U(f)$ _____ $L(f)$.

Proof.

□

Riemann Integrability

A bounded function f defined on $[a, b]$ is **Riemann-integrable** if _____.

If this is the case, then we define

$$\int_a^b f =$$

After Class:

Pick & Prove Problems:

★ Exercise 7.2.2

Consider $f(x) = 1/x$ over the interval $[1,4]$. Let P be the partition consisting of the points $\{1, 3/2, 2, 4\}$.

- (a) Compute $L(f, P)$, $U(f, P)$, and $U(f, P) - L(f, P)$.
- (b) What happens to the value of $U(f, P) - L(f, P)$ when we add 3 to the partition?
- (c) Find a partition P' of $[1,4]$ for which $U(f, P') - L(f, P') < 2/5$.

★ Exercise 7.2.4

Let g be bounded on $[a, b]$ and assume there exists a partition P with $L(g, P) = U(g, P)$. Describe the function g . Is it integrable? Explain why or why not.

★★ Exercise 7.2.1 (Not valid for HW because we did it in class)

Let f be a bounded function on $[a, b]$ and let P be an arbitrary partition of $[a, b]$. First, prove that $U(f) \geq L(f, P)$. Then, prove that $U(f) \geq L(f)$.

★★★ Exercise 7.2.6

Show that if f satisfies *Riemann's Original Definition of the Integral* (see exercise in book), then f is integrable in the sense of Definition 7.2.7.

Understanding Analysis Sections Covered: 7.2, 7.3

Definitions and Theorems to include in Analysis Journal:

- Definition 7.2.8 (Integrability criterion)
 - Theorem 7.2.9 (continuous $\rightarrow$ integrable)
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Before Class:

Readings: Sections 9.5 and 9.6 of Alcock.

Analysis Journal Prompts:

- Answer Alcock's parting questions in Section 9.5. Must a non-continuous function be non-integrable? How "weird" could a function be and still be integrable?
- Look back at the very first proof we did in class on the first day. How is Riemann's Criterion (Riemann's condition) related to this statement?

Riemann's Criterion

A bounded function f is integrable on $[a, b]$ if and only if...

Proof.

□

Example 1: Use Riemann's Criterion to show that $f(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x \neq 0 \end{cases}$ is integrable.

Theorem

If f is continuous on $[a, b]$, then it is integrable on $[a, b]$.

Proof.

Example 2: Topologist's Sine Function $f(x) = \begin{cases} x \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Example 3: Thomae's Function $t(x) = \begin{cases} 1 & \text{if } x = 0 \\ \frac{1}{n} & \text{if } x = \frac{m}{n} \in \mathbb{Q} \text{ (in lowest terms, } n > 0) \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$

Example 4: Cantor Function $f(x) = \begin{cases} 1 & \text{if } x \in C \\ 0 & \text{if } x \notin C \end{cases}$

After Class:

Pick & Prove Problems:

★ Exercise 7.3.1

Consider the function

$$h(x) = \begin{cases} 1 & \text{for } 0 \leq x < 1 \\ 2 & \text{for } x = 1 \end{cases}$$

over the interval $[0,1]$.

- (a) Show that $L(f, P) = 1$ for every partition P of $[0,1]$.
- (b) Construct a partition P for which $U(f, P) < 1 + 1/10$.
- (c) Given $\varepsilon > 0$, construct a partition P_ε for which $U(f, P_\varepsilon) < 1 + \varepsilon$.
- (d) Determine whether or not the function is integrable, and if so, compute the value of $\int_0^1 f$.

★★ Exercise 7.2.7

Let $f : [a, b] \rightarrow \mathbb{R}$ be increasing on the set $[a, b]$ (i.e. $f(x) \leq f(y)$ whenever $x < y$). Show that f is integrable on $[a, b]$.

★★ Exercise 7.3.3

Let

$$\begin{cases} 1 & \text{if } x = 1/n \text{ for some } n \in \mathbb{N} \\ 0 & \text{otherwise.} \end{cases}$$

Show that f is integrable on $[0,1]$ and compute $\int_0^1 f$.

★★ Supplemental Exercise

Prove that a bounded function defined on $[a, b]$ with a finite number of discontinuities in $[a, b]$ must be integrable on $[a, b]$.

★★ Exercise 7.3.7 (Modified)

- (a) Assume that $f : [a, b] \rightarrow \mathbb{R}$ is integrable. Show that if $g(x) = f(x)$ for all but one point in $[a, b]$, then g is integrable as well.
- (b) Provide an example to show that g may fail to be integrable if it differs from f at a countably infinite number of points.

★ ★ ★ Exercise 7.2.3 (Sequential Criterion for Integrability)

- (a) Prove that a bounded function f is integrable on $[a, b]$ if and only if there exists a sequence of partitions $(P_n)_{n=1}^\infty$ satisfying

$$\lim_{n \rightarrow \infty} [U(f, P_n) - L(f, P_n)] = 0$$

- (b) For each n , let P_n be the partition of $[0, 1]$ into n equal subintervals. Find formulas for $U(f, P_n)$ and $L(f, P_n)$ if $f(x) = x$. The combinatorial formula $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$ will be useful.
- (c) Use the sequential criterion for integrability from (a) to show directly that $f(x) = x$ is integrable on $[0, 1]$ and compute $\int_0^1 f$.

Understanding Analysis Sections Covered: 7.4 + Chapter 11.5 (Hunter, UCDavis)

This is the first time I have been unsatisfied with the textbook I chose. I don't think it goes about proving these in a logical way, in an order that makes sense, and to the specificity I think you would benefit from. I have provided you with the supplemental notes that I think go through it well. (See Moodle/Handout)

Definitions and Theorems to include in Analysis Journal:

- Theorem 11.32, 11.33 (Linearity, 7.4.2 ii) and i) in UA)
- Theorem 11.35 (product and quotient)
- Theorem 11.36 (monotonicity, 7.4.2 iv) in UA)
- Theorem 11.37 (7.4.2 iii) in UA)
- Theorem 11.38 (average value theorem)
- Theorem 11.40 (absolute value, 7.4.2 v) in UA)
- Theorem 11.44 (additivity, 7.4.1 in UA)
- Definition 11.45 (7.4.3 in UA)

Before Class:

Readings: Section 9.7, Alcock

Analysis Journal Prompts:

- How does this “Algebraic Integrability Theorem” feel similar to all of our other algebraic theorems in this course? How does it feel different?

Linearity

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be integrable functions. Then

$$\int_a^b cf =$$

$$\int_a^b (f + g) =$$

Proof.

Scratch Work (part b):

Product and Quotient

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be integrable functions. Then

Further, suppose _____ and _____. Then

Additivity

Let $f : [a, b] \rightarrow \mathbb{R}$ be integrable and $a < c < b$. Then

Proof Sketch:

Monotonicity

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be integrable functions and suppose $f \leq g$. Then

Proof.

□

Bounds on the Integral (Corollary)

Let $f : [a, b] \rightarrow \mathbb{R}$ be integrable and suppose $m \leq f \leq M$. Then

Proof.

□

Average Value Theorem

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then there exists $x \in [a, b]$ such that

$$f(x) = \frac{1}{b-a} \int_a^b f$$

Proof.

□

Absolute Value

Let $f : [a, b] \rightarrow \mathbb{R}$ be integrable. Then $|f|$ is integrable and

$$\left| \int_a^b f \right| \leq \int_a^b |f|$$

Before Class:***Readings:*** None***Analysis Journal Prompts:***

- Finish your Definition and Theorem map for these two units. Which proofs rely on which theorems/definitions?
- Come to class with an updated “Questions about Analysis” list. We’ll try to answer some in review for the test.

Before Class:***Readings:*** Section 9.8 and 9.9, Alcock***Analysis Journal Prompts:***

- Why is it called the “Fundamental” Theorem of Calculus? What’s so fundamental about it?

Fundamental Theorem of Calculus

1) If $f : [a, b] \rightarrow \mathbb{R}$ is integrable, and $F : [a, b] \rightarrow \mathbb{R}$ satisfies $F'(x) = f(x)$ for all $x \in [a, b]$, then

2) Let $g : [a, b] \rightarrow \mathbb{R}$ be integrable, and for $x \in [a, b]$, define

$$G(x) =$$

Then G is _____ on $[a, b]$.

If g is continuous at $c \in [a, b]$, then G is differentiable at c and _____.

Proof.

Some last questions to ponder:

1. Exercise 7.5.2 (True/False)

(a) If g is continuous on $[a, b]$, then $g = h'$ for some $h : [a, b] \rightarrow \mathbb{R}$.

(b) If $g = h'$ for $[a, b] \rightarrow \mathbb{R}$, then g is continuous on $[a, b]$.

2. *Are derivatives necessarily integrable?*